$r(T) = 3$

$\dim(U) = r(T) + n(T)$

$\Rightarrow 4 = 3 + n(T)$

$\Rightarrow n(T) = 1$

Q: $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$T(e_1) = e_1 + e_2 = (1,1,0)$

$T(e_2) = e_2 + e_3 = (0,1,1)$

$T(e_3) = e_1 + e_2 + e_3 = (1,1,1)$

$r(T) = 3$

$n(T) = 3 - 3 = 0$

$\therefore N(T) = \{0\}$

$\therefore T$ is 1-1

Check T is invertible or not. If yes then find T^{-1}.

$r(T) = \dim(\mathbb{R}^3)$

$\therefore T$ is onto.

Let $(x,y,z) \in \mathbb{R}^3$

$(x,y,z) = \alpha e_1 + \beta e_2 + \gamma e_3$

$\Rightarrow \alpha = x, \beta = y, \gamma = z$

$T(x,y,z) = T(xe_1 + ye_2 + ze_3)$

$= xT(e_1) + yT(e_2) + zT(e_3)$

$= x(e_1 + e_2) + y(e_2 + e_3) + z(e_1 + e_2 + e_3)$

$= ((x+z)e_1, (x+y+z)e_2, (y+z)e_3)$

$= (x+z, x+y+z, y+z)$

Put $T(x,y,z) = (0,0,0)$

$x + z = 0$

$x + y + z = 0$

$y + z = 0$

$\Rightarrow x = 0, y = 0, z = 0$

$\therefore N(T) = \{0\}$

$\therefore T$ is one-one

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